

1 MPPI Algorithm Pseudocode

Algorithm 1 Model Predictive Path Integral Control (MPPI)

Require: Current state x_t , dynamics model f , stage cost function c , terminal cost function c_f , number of samples N , control horizon T , noise covariance Σ_u , temperature parameter λ

Ensure: Optimal control sequence $\mathbf{u}^*$

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1: Initialization:
2:   Control sequence  $\mathbf{u}_0 = \mathbf{0} \in R^{T \times m}$  (or previous control sequence)
3:   Noise sequences  $\{\delta \mathbf{u}_i\}_{i=1}^N$ , where  $\delta u_i^k \sim \mathcal{N}(0, \Sigma_u), \forall k = 1, \dots, T$ 
4: Generate Sampled Trajectories:
5: for  $i = 1$  to  $N$  do
6:    $x_i^0 = x_t$ 
7:    $\mathbf{u}_i = \mathbf{u}_0 + \delta \mathbf{u}_i$ 
8:    $J_i = 0$ 
9:   for  $k = 0$  to  $T - 1$  do
10:     $x_i^{k+1} = f(x_i^k, u_i^k)$ 
11:     $J_i += c(x_i^k, u_i^k)$ 
12:   end for
13:    $J_i += c_f(x_i^T)$ 
14: end for
15: Calculate Weights:
16:    $J_{min} = \min\{J_1, J_2, \dots, J_N\}$ 
17:    $\omega_i = \exp\left(-\frac{J_i - J_{min}}{\lambda}\right), \quad i = 1, \dots, N$ 
18:    $\omega_i = \omega_i / \sum_{j=1}^N \omega_j$ 
19: Calculate Optimal Control Sequence:
20:    $\mathbf{u}^* = \mathbf{u}_0 + \sum_{i=1}^N \omega_i \cdot \delta \mathbf{u}_i$ 
21: return  $\mathbf{u}^*$ 

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1.1 Notation

- x_t : Current system state
- $f(x, u)$: System dynamics model
- $c(x, u)$: Stage cost function
- $c_f(x)$: Terminal cost function
- N : Number of sampled trajectories
- T : Control horizon length
- Σ_u : Control noise covariance matrix
- λ : Weight temperature parameter

- $\mathbf{u}_0$: Base control sequence
- $\delta \mathbf{u}_i$: i -th noise sequence
- J_i : Total cost of i -th trajectory
- ω_i : Weight of i -th trajectory